

Zerin Shaima Meem

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SKILLS

Python, C++, JavaScript, TypeScript, SQL (MySQL), NoSQL (MongoDB, Supabase), Keras, TensorFlow, scikit-learn, NumPy, pandas, React, Angular, Node.js, Git/GitHub, Google Colab, Postman, Jest, Storybook, Playwright

EDUCATION

Masters in Computer Science — AI-Concentration — Expected May 2027 **CGPA :3.89/4.00**

Relevant Coursework: AI, Adv. AI, Deep Learning, Cloud Computing, Algorithms, Software Engineering

WORK EXPERIENCE

- **Graduate Research Assistant at University of Nebraska Omaha** **August 2025- Present**
Advisor: Dr. Xiaoqian (Tiffany) Zhang, Assistant Professor of Computer Science
 - Investigated and evaluated rate adaptation algorithms using NS-3 simulations; analyzed experimental outputs to compare network performance under different traffic scenarios.
Funded by the Nebraska Research Initiative Award 2025-2026.
 - Designed and developed AI-driven congestion control experiments for large-scale storage systems, focusing on performance metrics, result analysis, and model-driven decision making.
Funded by NSF CRII award 2025-2027.
- **Software Engineer I at Samsung R&D Institute Bangladesh** **January 2025- July 2025**
 - Collaborated on a web-based semantic knowledge graph modeling tool that enables ontology creation, visualization, and data mapping. Built a persistent **real-time graph canvas** with **drag-and-drop** support, allowing users to visually construct and edit RDF-based knowledge structures. Took part in unit testing, end-to-end test automation, UI component isolation and clean delivery through Git-based workflows.

PROJECTS

- 🔗 **Portfolio Builder** *React, Node.js, AWS RDS/PostgreSQL, AWS Amplify, JavaScript*
 - A Full-stack web application that allows users to create accounts, customize personal portfolio pages, and publish shareable portfolio links.
 - Integrated relational data storage using AWS RDS/PostgreSQL to manage user credentials, profile details, layout configurations, and customization data; supported responsive UI development, authentication workflows, backend APIs, and GitHub collaboration.
- 🔗 **CareTrack-Connect** *JavaScript, Node.js, Supabase, SQL, Vite*
 - Built a digital health tracking application that enables patients to submit daily wellness questionnaire responses and allows doctors to review patient data before and after appointments.
 - Developed backend for storing and retrieving information of patients, doctors, and questionnaire responses using API and database workflows. Worked with project documentation including API reference and database schema and strengthened experience in structured data, backend services, and team-based software delivery.
- 🔗 **CUET Competitive Programming Community Management System** *MEAN stack*
 - A MEAN stack web application for monitoring programming community members with real-time leaderboard generation. Uses **API-based data extraction** from Codeforces, AtCoder, CodeChef, and other competitive programming platforms. Built performance visualization features for 130 competitive programmers at CUET.

PUBLICATION

- 🔗 **A CSI-based Position Independent Gesture Recognition System Using DL** *IEEE WIECON-ECE 2024*
 - An LSTM-based classifier achieving **81.1%** accuracy for position independent gesture recognition using WiFi CSI data. Designed data pipeline: ESP-32 CSI data collection → preprocessing → ML/DL model evaluation.
- 🔗 **Spot Light: Disaster Safety Assistance Mobile Application** *IEEE ICCPCT 2024*
 - An application with emergency messaging, fund transfers, rescue center mapping. Evaluated by users, with **52%** rating it highly effective in delivering disaster assistance.

ACHIEVEMENTS

- **GRACA Scholarship (\$5,000)**, UNO, Summer 2026, for project titled “Adaptive Rate Control for UAV Communication Networks Using Deep Reinforcement Learning.”
- **Honorable Mention**, UNO C-Bytes Research Showcase Competition.
- Solved 3000+ programming problems | 🏆Codeforces- **1607(Expert)** | 🏆CodeChef- **1788(3*)** | 3-time ICPC Dhaka Regional participant | **Globally ranked 9th** in ICPC AlgoQueen 2024